Show that a single-mode thermal state of light is first-order coherent but not second- or higher-order coherent.

顯示思路

To understand the coherence of a thermal state, we need to look at the statistical properties of the light field using the **normalized correlation functions** $g^{(n)}$. We'll explore this together by deriving these values step-by-step. 💡

In quantum optics, a single-mode thermal state (like light from a star or a blackbody) is described by a density operator $\hat{\rho}$ where the photon number follows the **Bose-Einstein distribution**:

$$P(n) = \frac{\bar{n}^n}{(1+\bar{n})^{n+1}}$$

Here, $\bar{n}$ is the average number of photons in the mode. To "show" coherence properties, we must calculate the expectation values of the creation ($\hat{a}^\dagger$) and annihilation ($\hat{a}$) operators.

The Correlation Functions

The n-order coherence function at zero time delay ($\tau = 0$) for a single mode is generally defined as:

$$g^{(n)}(0) = \frac{\langle (\hat{a}^\dagger)^n \hat{a}^n \rangle}{\langle \hat{a}^\dagger \hat{a} \rangle^n}$$

For a state to be considered "fully coherent," it must satisfy $|g^{(n)}(0)| = 1$ for all n. If $g^{(1)}(0) = 1$, the light can produce perfect interference fringes in a Michelson interferometer, but that doesn't tell the whole story about its fluctuations. 🌊

Step 1: First-Order Coherence

To begin, let's look at the first-order coherence function $g^{(1)}(0)$. By definition:

$$g^{(1)}(0) = \frac{\langle \hat{a}^\dagger \hat{a} \rangle}{\langle \hat{a}^\dagger \hat{a} \rangle}$$

Based on this definition, what is the numerical value of $g^{(1)}(0)$ for **any** state with a non-zero average photon number?

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思考型

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